

Data Publication

Introduction to Data Management Practices course

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Why submit to a repository?

“The data is available upon request”

Many reasons:

- Open Science & FAIR
- Reproducibility
- Trail of evidence
- 3rd party access
- Archival purposes
- Publication of paper requires it



Digitalbevaring.dk

Credit: Illustration from Digitalbevaring.dk / Jørgen Stamp (CC BY 2.5 Denmark license).

Why submit to a repository?

Data publication is the best way to make your research projects FAIR since your data becomes:

- **Findable** by being assigned a persistent identifier, and by being described with rich metadata
- **Accessible** by being put in a resource that is searchable, and enables easy access via internet
- **Interoperable** by using standard format and language to represent both the data and its metadata
- **Reusable** by fulfilling the F, A, and I, and by having a clear and accessible data usage license

What is data?

What research outputs should be submitted?

- **Raw data:** straight from the instrument eg fastq, bam, cram
- **Processed data:** normalization, removal of outliers, expression measurements, statistics
- **Metadata:** minimum information to reproduce the data, sample information, precise protocols

Types of repositories

- **Domain-specific:**
 - Best choice - long-term plan, typically free, maximum reach
 - E.g. [European Nucleotide Archive](#), [European Genome Phenome Archive](#), [ArrayExpress](#), [PRIDE](#)
- **General purpose:**
 - Second best - long-term plan, might cost (now or in future), good reach but less specific in metadata → more difficult for future users to judge if a dataset will be useful
 - E.g. [Zenodo](#), [\(SciLifeLab\) Figshare](#), [Dryad](#)
- **In-house/institutional**
 - For archive/backup purpose mainly, might cost, limited reach unless also published in a data catalogue

Domain-specific

General
purpose

In-house

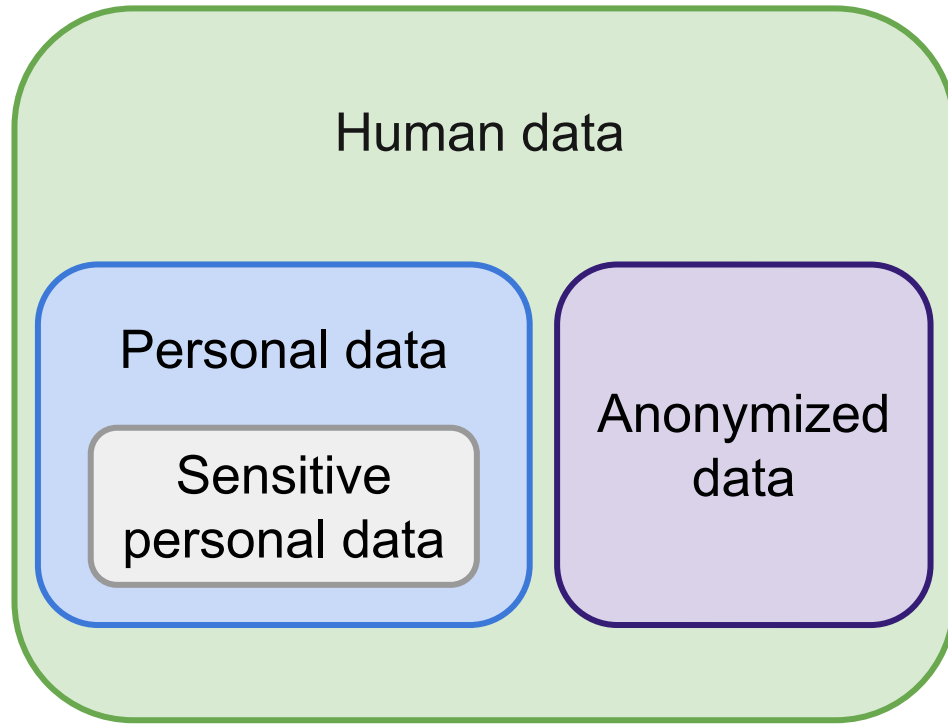
Repositories for sensitive personal data

European Genome-Phenome Archive (EGA) is a repository for all types of personally identifiable genetic and phenotypic data resulting from biomedical research projects.

- Hosted by EMBL-EBI and CRG
- Launched 2008
- Legal challenges for Swedish universities



Sensitive personal data



Human data is any data associated with or derived from a human being.

Personal data is any information that refers to an identified or identifiable living person.

Anonymized data is personal data that has been stripped of any personally identifiable information.

Sensitive personal data is special categories of personal data that need extra protection.

What data is regarded as personal data?

Not personal data

Name:
'Eva Johansson'

Probably not
personal data

Name:
'Eva Johansson'

City:
'Malmö'

Probably personal data

Name:
'Eva Johansson'

City:
'Malmö'

Date of birth:
'1985-01-01'

Data may or may not be regarded as
personal data depending on the context

What is sensitive personal data?

Special categories of personal data defined by GDPR (Art. 9)

Health Data

Sex life or
Sexual orientation

Racial or
Ethnic Origin

Biometric
Data

Genetic
Data

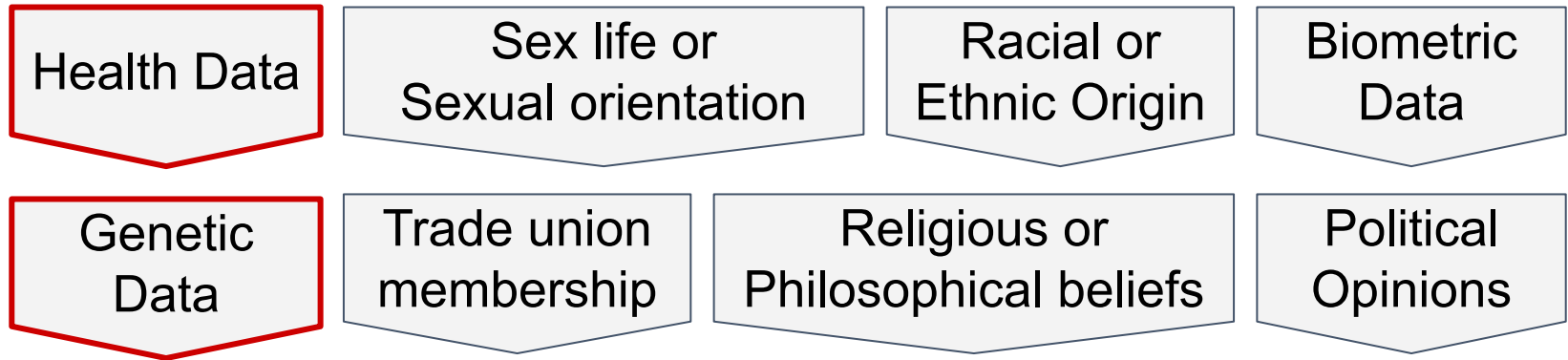
Trade union
membership

Religious or
Philosophical beliefs

Political
Opinions

What is sensitive personal data?

Special categories of personal data defined by GDPR (Art. 9)



Processing of sensitive personal data is prohibited, except under certain circumstances (e.g. research).

When personal data becomes sensitive

Personal data

Name:
'Eva Johansson'

City: 'Malmö'

Date of birth:
'1985-01-01'



Not personal
data

Diagnosis:
'diabetes, type 1'



Sensitive
personal data

Name:
'Eva Johansson'

City: 'Malmö'

Date of birth:
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Repositories for sensitive personal data

Federated EGA (FEGA) is a network of national human data repositories in collaboration with the central EGA.

- FEGA Sweden is in production since February 2024
- Hosted by NBIS/UU
- Simplify the legal situation
- Provide support and services adapted to the Swedish life science research community



Evaluate a repository

Things to check when evaluating:

- Are others in the community using it?
- Is it easy to navigate / user-friendly?
- Is there support / guidance for submission and reuse?
- Is it sustainable, i.e. will the repository be around for a while?
- Will the datasets obtain persistent identifiers? Is the repository itself FAIR?

Identify repositories

How to find a suitable repository for life science data?

- [EBI repository wizard](#) - guide depending on data type
- [ELIXIR deposition databases](#) - core resources with long-term data preservation and accessibility plans
- [FAIRsharing.org/databases](#) - catalogue of many repositories, with possibility to filter on e.g. domain
- [Scientific Data Repository Guidance](#) - publisher's recommendation
- [re3data.org](#) - registry of research data repositories (not only life science)

Demo: EBI Repository Wizard

Which repository would be suitable if you have a genomics project with mice RNA sequences?

- Go to <https://www.ebi.ac.uk/submission/>
- Answer the questions regarding
 - data type
 - need for controlled access
 - if experimentally produced by you
 - type of study

Demo: EBI Repository Wizard

Which repository would be suitable if you have a genomics project with mice RNA sequences?

- Go to <https://www.ebi.ac.uk/submission/>
- Answer the questions regarding
 - data type (DNA/RNA sequence)
 - need for controlled access (No)
 - if experimentally produced by you (Yes)
 - type of study (Other)
- Solution: [European Nucleotide Archive \(ENA\)](#)

Key Points

- Publishing data greatly increases the FAIRness of your research.
- Benefits of sharing data are several e.g. reproducibility purposes, follow the Open Science directive, meet requirement from publishers.
- If possible, use a domain-specific repository since it has maximum reach in the research community.
- The research output data types determines which domain-specific repository is suitable.